

12 (**long**) A building is being modelled for its response to an earthquake. A schematic of the building is shown in Fig. 9 consisting of a system of two spring coupled masses coupled to the ground by another spring as shown. The value of each mass m is 10^5 kg and the value of the spring constant k is 2×10^5 N m $^{-1}$. In response to an earthquake the ground displacement is $y = y_0 \cos \omega t$ where ω is the frequency of ground oscillations and y_0 is the ground displacement amplitude.

- (a) Show that the equation of motion for this system can be written in the form: [6]

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} 3k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2ky \\ 0 \end{bmatrix}$$

- (b) Determine the natural frequencies and mode shapes for this system. [8]

- (c) Determine expressions for the frequency response of the displacements of the two masses and sketch the frequency response for both masses separately. Show on your sketch the impact of added damping. [10]

- (d) During an earthquake the ground oscillates at an amplitude $y_0 = 1$ cm at an angular frequency $\omega = 1$ rad s $^{-1}$. Determine the displacement amplitudes of the masses for this case. [6]

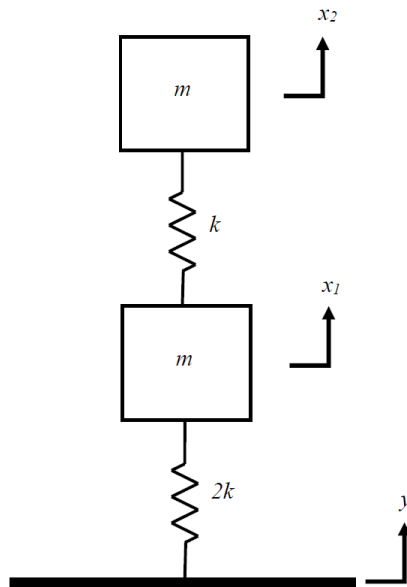


Fig. 9

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